

```

fit_data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\Sym4D_cutsAndFits';
%fit_data_file = 'FitEn_cut200_1Dsbg_dirGH_dE20_fix_bg_slope.mat'; % fit 1D
background with no Q-slope
fit_data_file =
'FitEn_cut200ref200_2D_constFFCut_dirGH_dE20_fix_bg_slope.mat';%Externaly 2-
exponent fit fixed bg at high momentum.
%fit_data_file = 'FitEn_cut200ref110_2D_dirGH_dE20_fix_bg_slope.mat';%Externaly
2-exponent fit fixed bg at high momentum.

if ~exist('fit_resEi200','var')
    ld = load(fullfile(fit_data_path,fit_data_file));
    if isfield(ld,'fit_resEi200')
        fit_resEi200 = ld.fit_resEi200;
    else
        fit_resEi200 = ld.fit_res;
    end
end
data_path = 'e:\SHARE\Fe\Data\analysis\06_fit_with_J0\sym4D_cutsAndFits';
fit_file = 'multicuts_fit_dataGH_ei200meV.mat';
if ~exist('cuts2fit200','var')
    ld = load(fullfile(data_path,fit_file));
    cuts2fit200 = ld.cuts2fit200;
end
cuts2fit = cuts2fit200;
fit_res = fit_resEi200;
mi = maps_instrument(200,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
%cl = {cuts2fit200.cut110_sel};
%cl = {cuts2fit200.cut200_sel};
cl = {cut(cuts2fit200.cut200_sel,[0,0.02,1],[])};

*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 4690699 pixels -- processing data... -----> retained 4604422 pixels
-----
Number of points in input object: 8259853
Fraction of object selected: 56.79% (= 4690699 points) ; Access speed: 1.87GB/sec
Fraction of object retained: 55.74% (= 4604422 points)

**** Total time in cut(sqw) : 1.0sec Including:
Data access time fraction: 17.7%; Transf. time frac.: 50.6%; Resulting pix preparation: 31.6%
-----

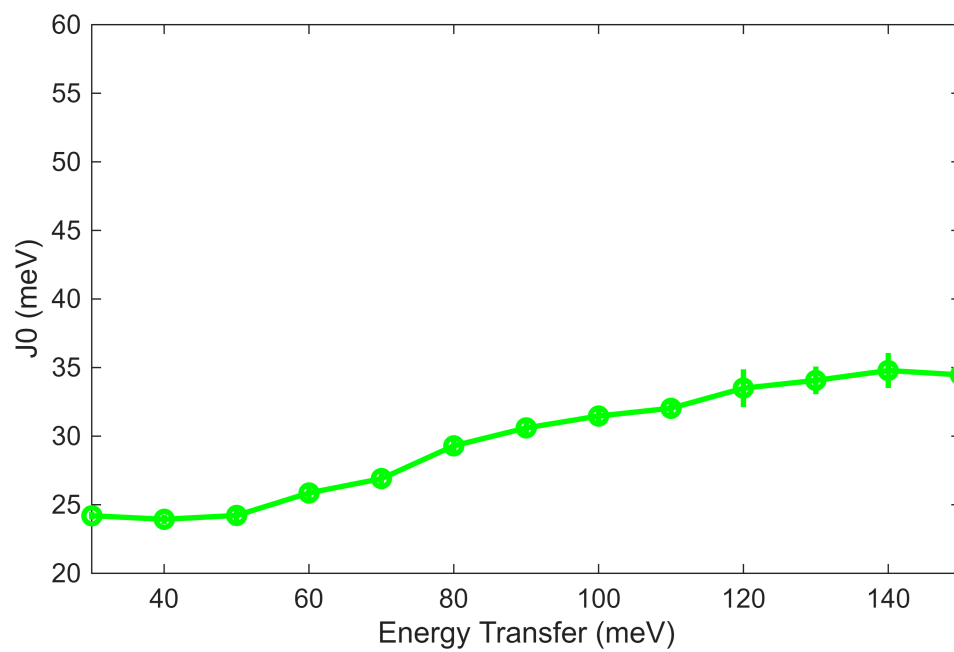
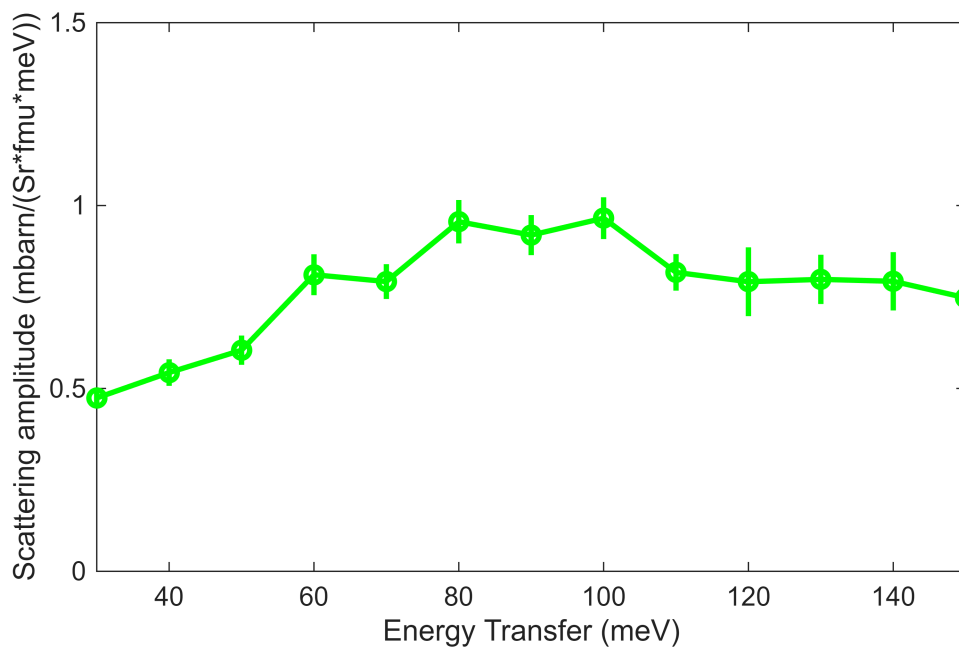
for i=1:numel(cl)
    cl{i} = cl{i}.set_instrument(mi);
    cl{i} = cl{i}.set_sample(sample);

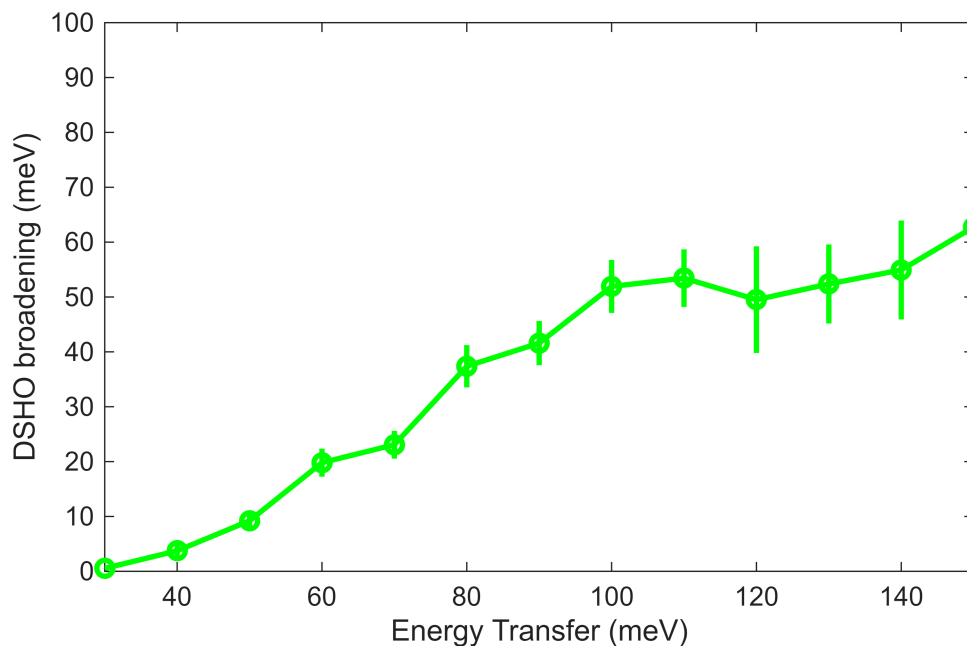
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_instrument(mi);
    %cuts2fit.cutsS_list{i} = cuts2fit.cutsS_list{i}.set_sample(sample);
end

the2Dcuts = cl;

[Sgn1,J0,G] = plot_SJG_results(fit_res,'g',{ 'S', 'J0', 'gamma'},{[0,1.5],[20,60],
[0,100]});

```





```
dE = min(Sgn1.x(2:end)-Sgn1.x(1:end-1));
e_min = Sgn1.x(1);
```

```
En = 90
```

```
En =
90
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
half_dE = 10;
en_idx = floor((En-e_min)/dE)+1;
%the2Dcuts = cuts2fit.cutsS_list;%(1:2);
%the2Dcuts = {cut(cuts2fit.cut110_sel,[0,0.02,2],[])};%(1:2);

%init_bg_params =
%cellfun(@(par)[par(1),par(3)],init_bg_params,'UniformOutput',false); % bg
%in case of 2-D fitting
%
[fit_obj,fit_par,figa,figb]=fit_single_set(the2Dcuts,En,half_dE,dE_step,init_fg_p
arams,init_bg_params,false)
lfit_par = fit_res.all_fit_par(en_idx);
en_range = lfit_par.en_range;
[fit_obj,fit_par,figa,figb]=refit_singleset_peaks(the2Dcuts,en_range,lfit_par,0.2
,true,false);
```

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.71301

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
 *** Step 1 of 1; Read data for 449588 pixels -- processing data... -----> included 405176 pixels

```
-----
Number of points in input object: 449588
Fraction of object selected: 100.00% (= 449588 points) ; Access speed: 1.14GB/sec
Fraction of object retained: 90.12% (= 405176 points)
```

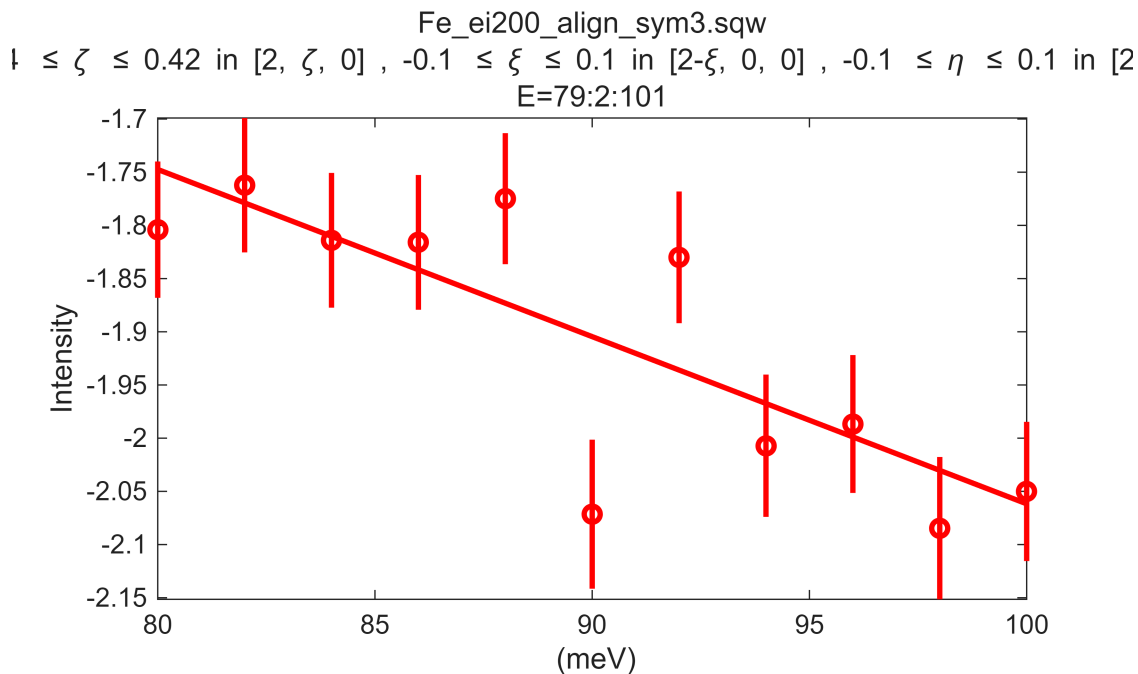
```

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 60.8%; Transf. time frac.: 37.7%; Resulting pix preparation:  0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for  148916 pixels -- processing data... -----> retained   139254 pixels
-----
Number of points in input object: 452915
      Fraction of object selected:   32.88% (=   148916 points) ; Access speed:      1.23GB/sec
      Fraction of object retained:   30.75% (=   139254 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 26.1%; Transf. time frac.: 43.7%; Resulting pix preparation: 30.1%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for  139254 pixels -- processing data... -----> included   139254 pixels
-----
Number of points in input object: 139254
      Fraction of object selected:  100.00% (=   139254 points) ; Access speed:      1.26GB/sec
      Fraction of object retained:  100.00% (=   139254 points)

**** Total time in cut(sqw)   :      0.0sec Including:
      Data access time fraction: 46.0%; Transf. time frac.: 50.2%; Resulting pix preparation:  0.0%
-----

```



```

-----
Beginning fit (max 20 iterations)-----
Starting point
-----
Total time = 2.3861s    Reduced Chi^2 = 15.2903
Free parameter values:
      41.6           0.9192           30.6           0.6125
-----
Iteration = 1
-----
Total time = 9.095s    Reduced Chi^2 = 1.3824    Levenberg-Marquardt = 0.1
Free parameter values:
      32.15          0.6222           29.34           0.2298
-----
Iteration = 2
-----
Total time = 15.6203s   Reduced Chi^2 = 1.337    Levenberg-Marquardt = 0.01
Free parameter values:
      35.53          0.7707           29.55           0.1646
-----
Iteration = 3

```

```

-----
Total time = 22.1799s    Reduced Chi^2 = 1.3363    Levenberg-Marquardt = 0.001
Free parameter values:
      38.13      0.8368      29.77      0.1455
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
  1          1
  2          8
  3      38.13 +/- 10.51
  4      0.8368 +/- 0.2241
  5          0
  6      29.77 +/- 1.15
  7          0
  8          0
  9          0
 10          0

Background parameter values:
-----
  1      0.1455 +/- 0.06247
  2     -0.01571
  3          0

Covariance matrix for free parameters:
-----
  1.0000  0.9707  0.9038 -0.9138
  0.9707  1.0000  0.8790 -0.9719
  0.9038  0.8790  1.0000 -0.8110
 -0.9138 -0.9719 -0.8110  1.0000

*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 452915 pixels -- processing data... -----> included 411177 pixels
-----
Number of points in input object: 452915
  Fraction of object selected: 100.00% (= 452915 points) ; Access speed: 1.90GB/sec
  Fraction of object retained: 90.78% (= 411177 points)

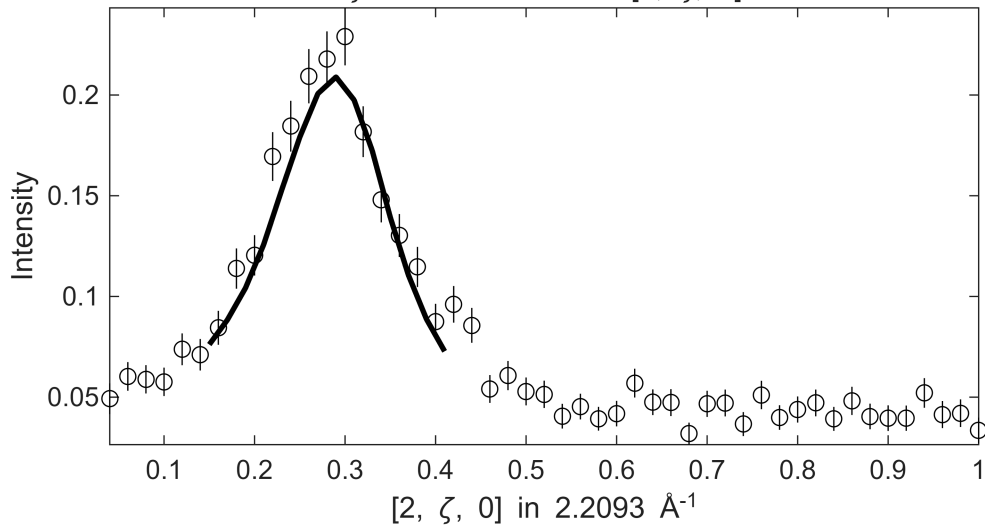
**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 57.0%; Transf. time frac.: 41.2%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; No resulting pixels requested
*** Step 1 of 1; Read data for 139254 pixels -- processing data... -----> included 126284 pixels
-----
Number of points in input object: 139254
  Fraction of object selected: 100.00% (= 139254 points) ; Access speed: 1.72GB/sec
  Fraction of object retained: 90.69% (= 126284 points)

**** Total time in cut(sqw) : 0.0sec Including:
  Data access time fraction: 40.3%; Transf. time frac.: 56.1%; Resulting pix preparation: 0.0%
-----
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for 452915 pixels -- processing data... -----> retained 411177 pixels
-----
Number of points in input object: 452915
  Fraction of object selected: 100.00% (= 452915 points) ; Access speed: 1.85GB/sec
  Fraction of object retained: 90.78% (= 411177 points)

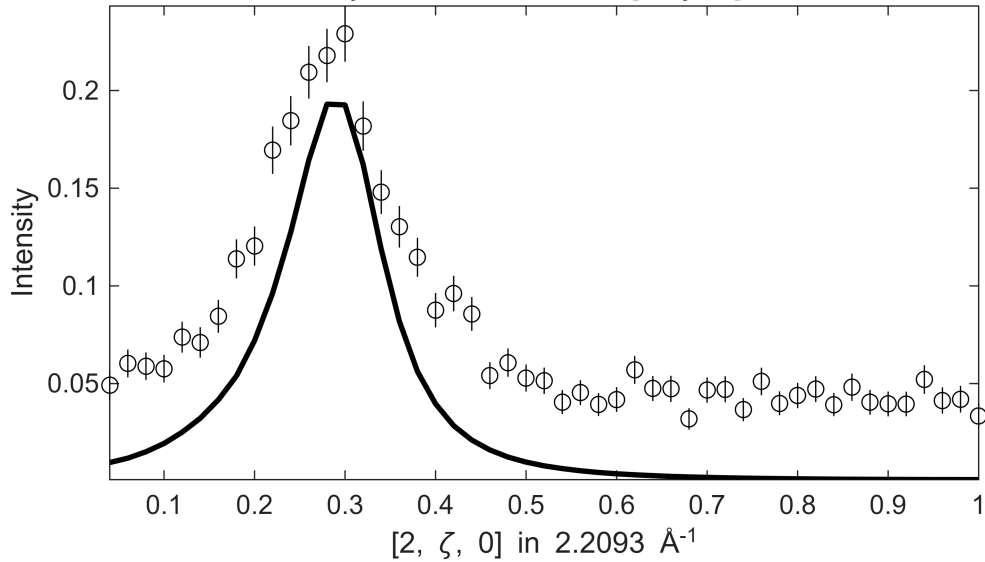
**** Total time in cut(sqw) : 0.1sec Including:
  Data access time fraction: 24.0%; Transf. time frac.: 51.7%; Resulting pix preparation: 23.5%
-----

```

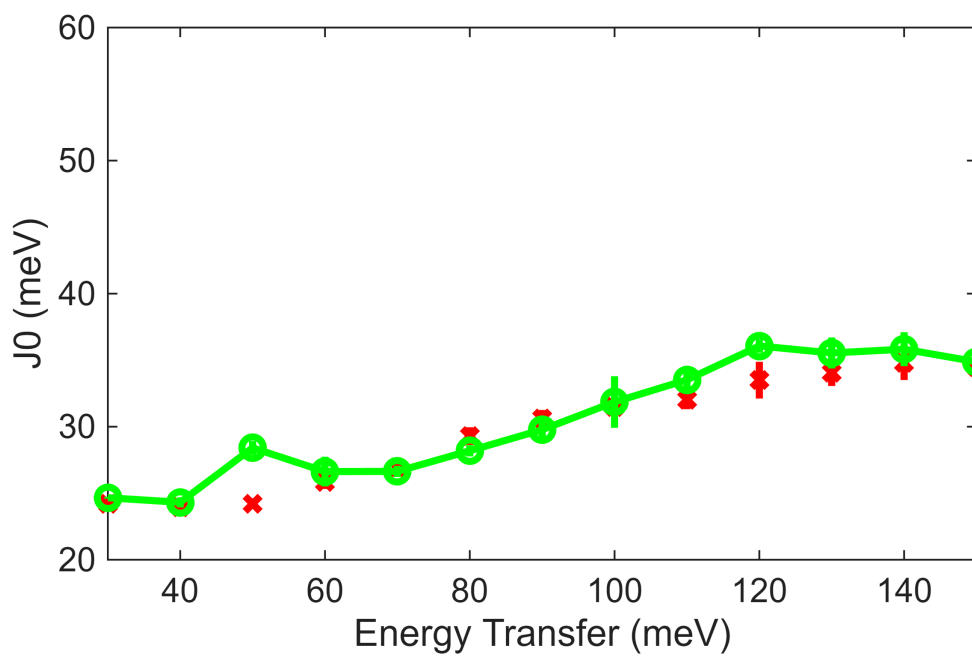
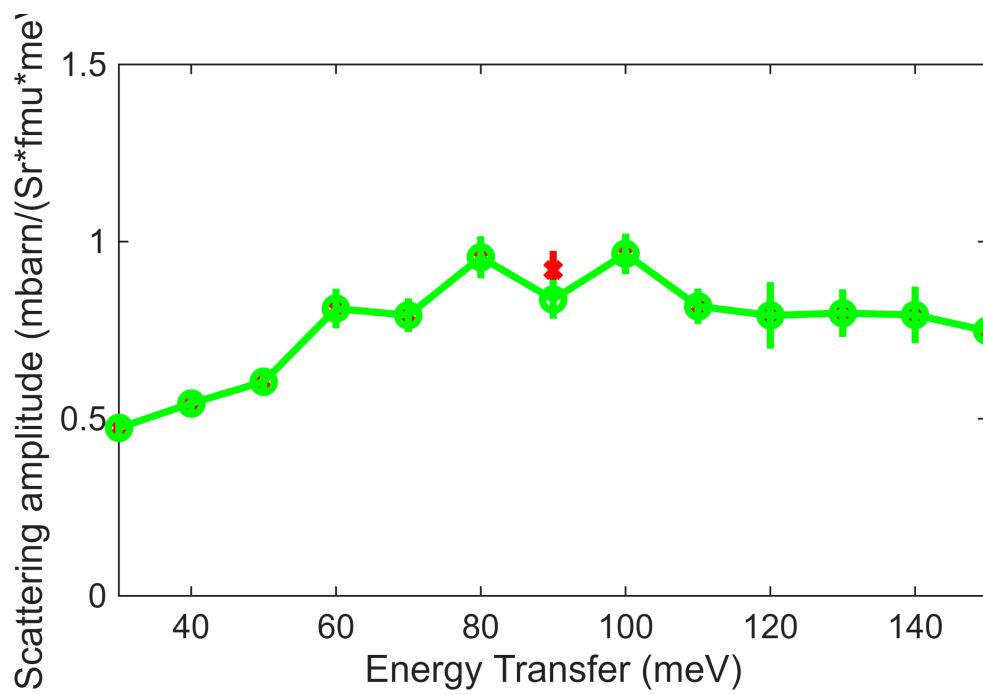
Fe_ei200_align_sym3.sqw
 $S=0.84\pm0.22$; $J0=30\pm1.1$; $\gamma=38.1291\pm10.512$
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$

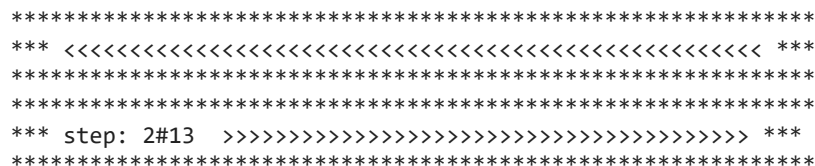


Fe_ei200_align_sym3.sqw
 $-0.1 \leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $80 \leq E \leq 100$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$



```
%fit_res.all_fit_par(en_idx) = fit_par;
%fit_res.all_fit_par(en_idx).p = fit_par.p;
Sgn1.signal(en_idx) = fit_par.p(4);
J0.signal(en_idx) = fit_par.p(6);
G.signal(en_idx) = fit_par.p(3);
plot_SJG_results(fit_res, 'g', {'S', 'J0', 'gamma'}, {[0,1.5],[20,60],[0,100]}),
{Sgn1,J0,G});
```



$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] \text{ , } \eta \in [-0.1, 0.1] \text{ in } [2, 0, \eta] \text{ , } 20 \leq \zeta \leq 200 \text{ in } [2, \zeta, 0]$$


S=0.34±0.022; J0= 24±0.045; gamma=-0.11698±0.00521833
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $30 \leq \zeta \leq 31$ in $[2, \zeta, 0]$



$$\begin{aligned} & \leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0] \text{ , } -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] \text{ , } 30 \leq \xi \\ & \zeta = -0.01:0.02:1.01 \text{ in } [2, \zeta, 0] \end{aligned}$$

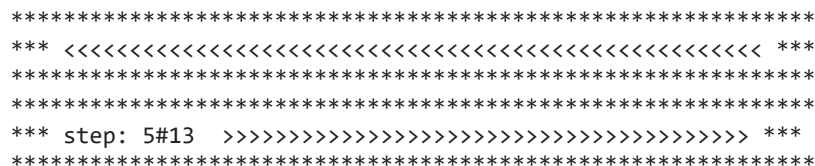
$$S=0.55\pm 0.034; J_0=24\pm 0.42; \gamma=9.18976\pm 0.302528$$

$$\leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 40 \leq E$$

$$\zeta=-0.01:0.02:1.01 \text{ in } [2, \zeta, 0]$$


$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 40 \leq \zeta \leq 60 \text{ in } [2, \zeta, 0]$$


S=0.67±0.16; J0= 26±0.57; gamma=16.2301±5.0446
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $50 \leq E$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$

$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 50 \leq \zeta \leq 100 \text{ in } [2, \zeta, 0]$$


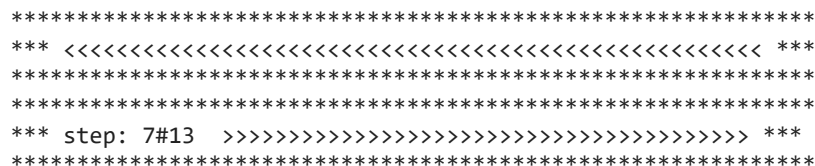
S=0.39±0.021; J0= 26±0.43; gamma=8.60482±0.352268
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $60 \leq E$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$



$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 60 \leq \zeta \leq 100 \text{ in } [2, \zeta, 0]$$


S=0.54±0.033; J0= 28±0.41; gamma=18.7887±0.694564
 $\leq \xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $70 \leq E$
 $\zeta=-0.01:0.02:1.01$ in $[2, \zeta, 0]$



$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, \bar{0}] , -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta] , 70 \leq \zeta \leq 100 \text{ in } [2, \zeta, 0]$$


Fe ei200 align sym3.sqw

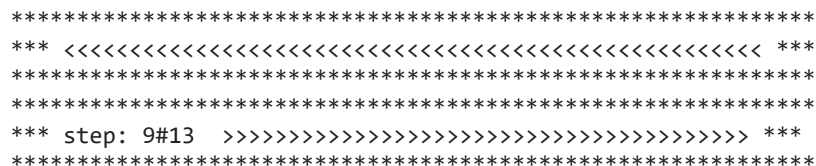
$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 80 \leq E$$

$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 80 \leq E$$


Fe ei200 align sym3.sqw

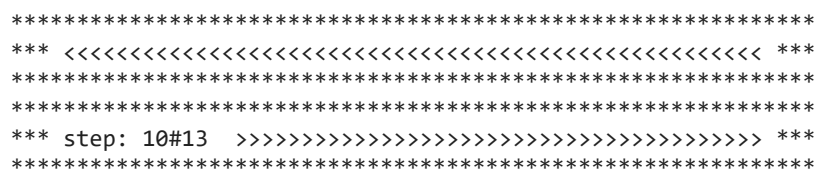
$$\leq \xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 90 \leq E$$

$$\zeta = -0.01:0.02:1.01 \text{ in } [2, \zeta, 0]$$

$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 90 \leq E$$


Fe ei200 align sym3.sqw

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 100 \leq \zeta \leq 1000 \text{ in } [2, \zeta, 0]$$

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 100 \leq \zeta \leq 1000 \text{ in } [2, \zeta, 0]$$


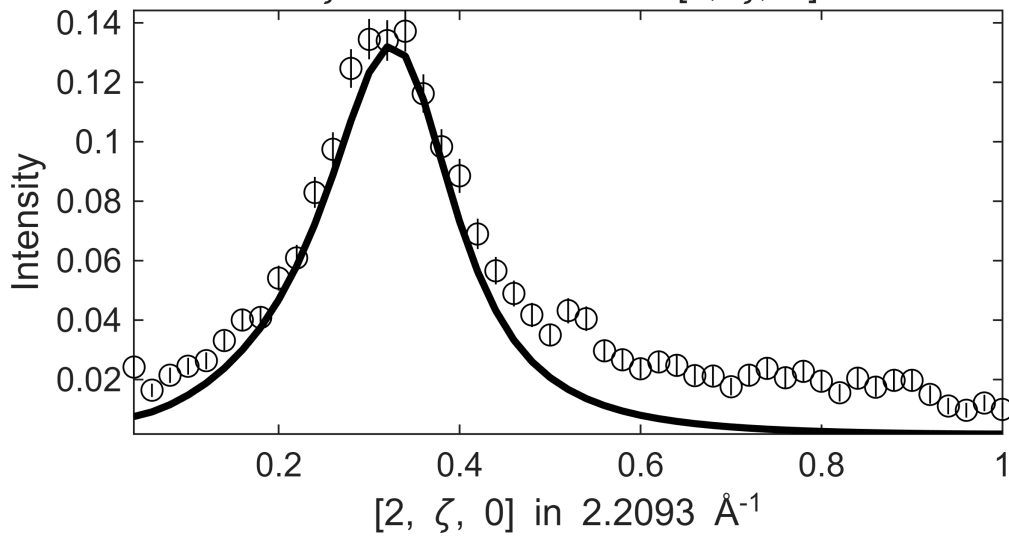
Fe ei200 align sym3.sqw

$$\xi \in [-0.1, 0.1] \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 110 \leq E$$

$$\zeta = -0.01:0.02:1.01 \text{ in } [2, \zeta, 0]$$

Fe_ei200_align_sym3.sqw

$\xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $110 \leq E$
 $\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



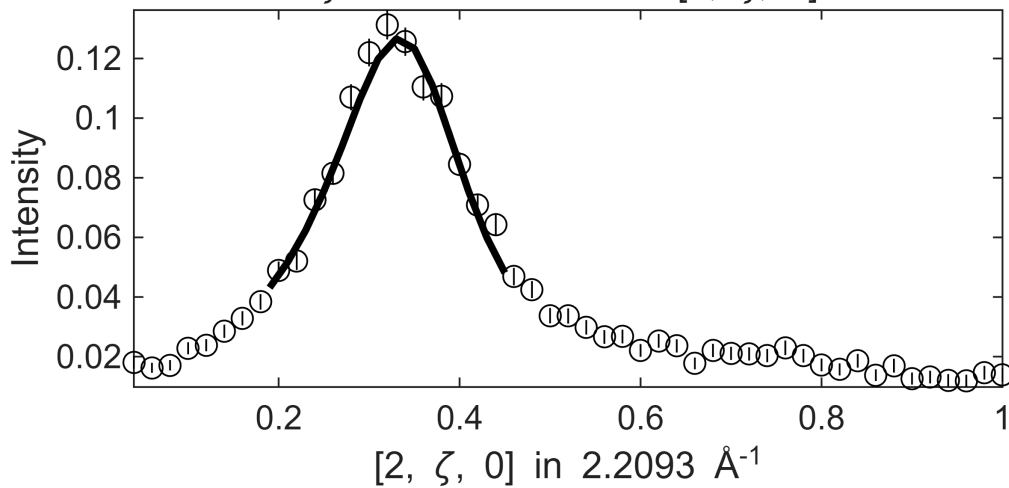
```
*****
*** <<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<<< ***
*****
*****
*** step: 11#13 >>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>>> ***
*****
```

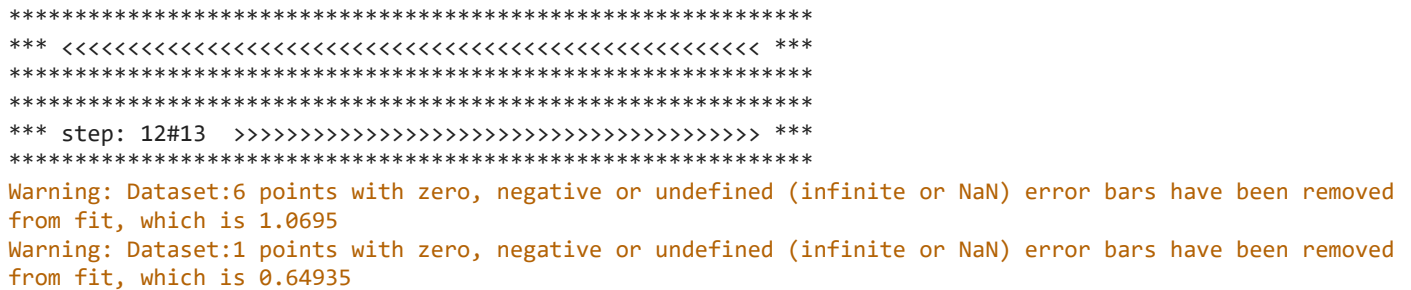
Warning: Dataset:7 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.2478

Fe_ei200_align_sym3.sqw

$S=0.95 \pm 0.17$; $J_0 = 35 \pm 1.1$; $\gamma = 73.4147 \pm 11.5496$

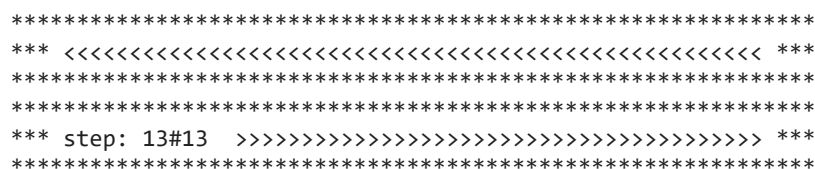
$\xi \leq 0.1$ in $[2-\xi, 0, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, \eta]$, $120 \leq E$
 $\zeta = -0.01:0.02:1.01$ in $[2, \zeta, 0]$



$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 120 \leq \zeta \leq 180 \text{ in } [2, \zeta, 0]$$


$S=0.97\pm0.21$; $J_0=36\pm 1.3$; $\gamma=77.5312\pm14.3801$

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 130 \leq \zeta \leq 150 \text{ in } [2, \zeta, 0]$$

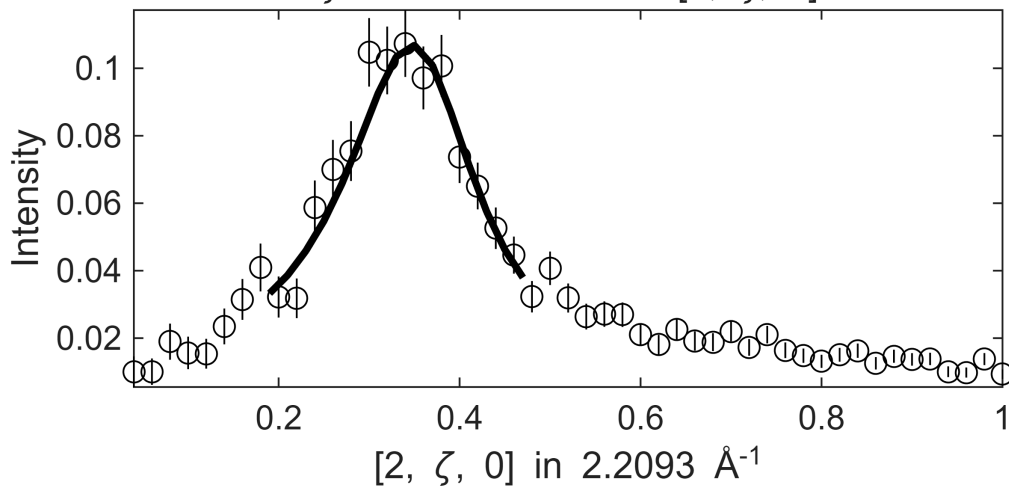
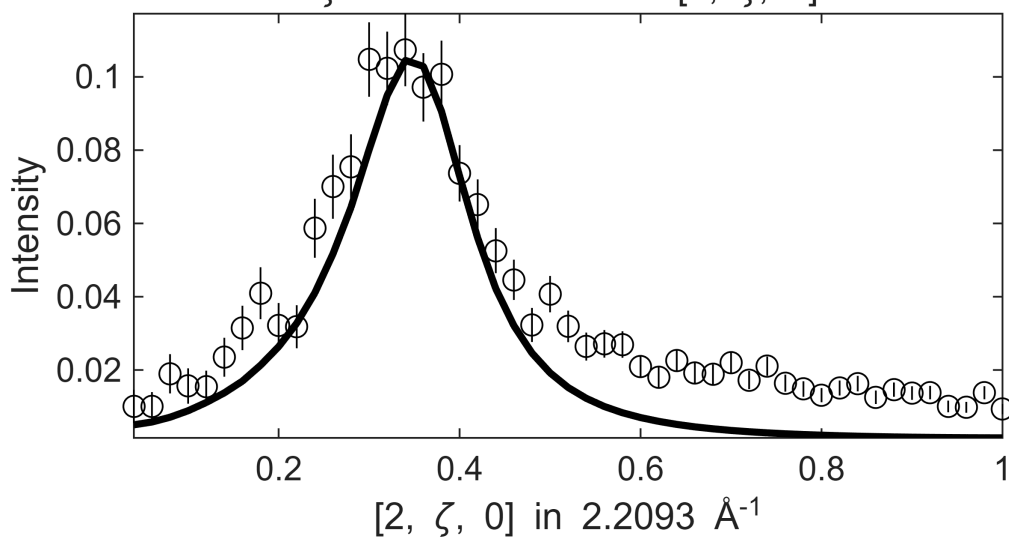

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 130 \leq \zeta \leq 140 \text{ in } [2, \zeta, 0]$$


Fe ei200 align sym3.sqw

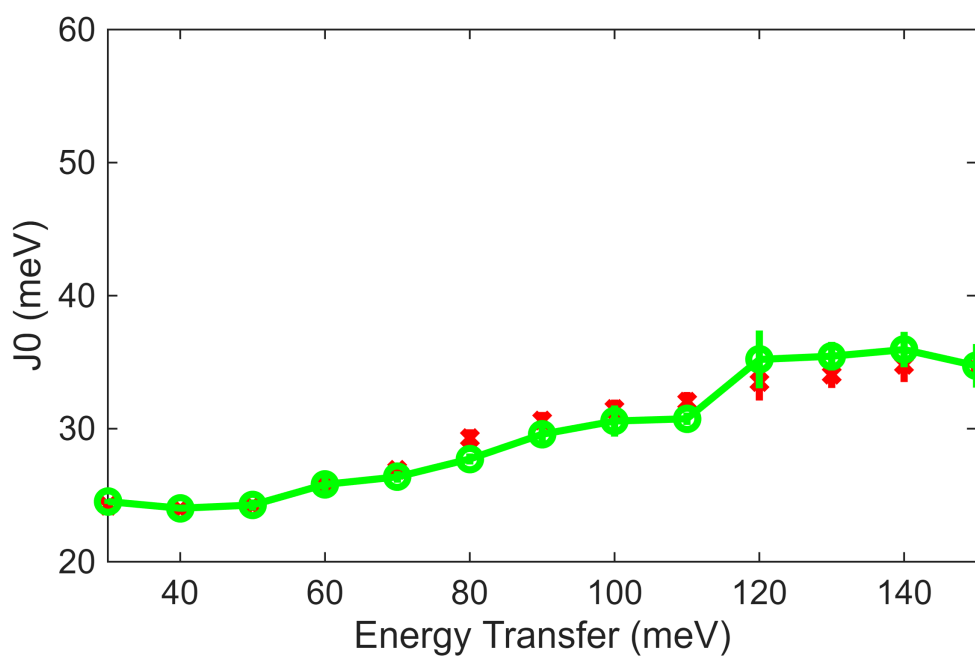
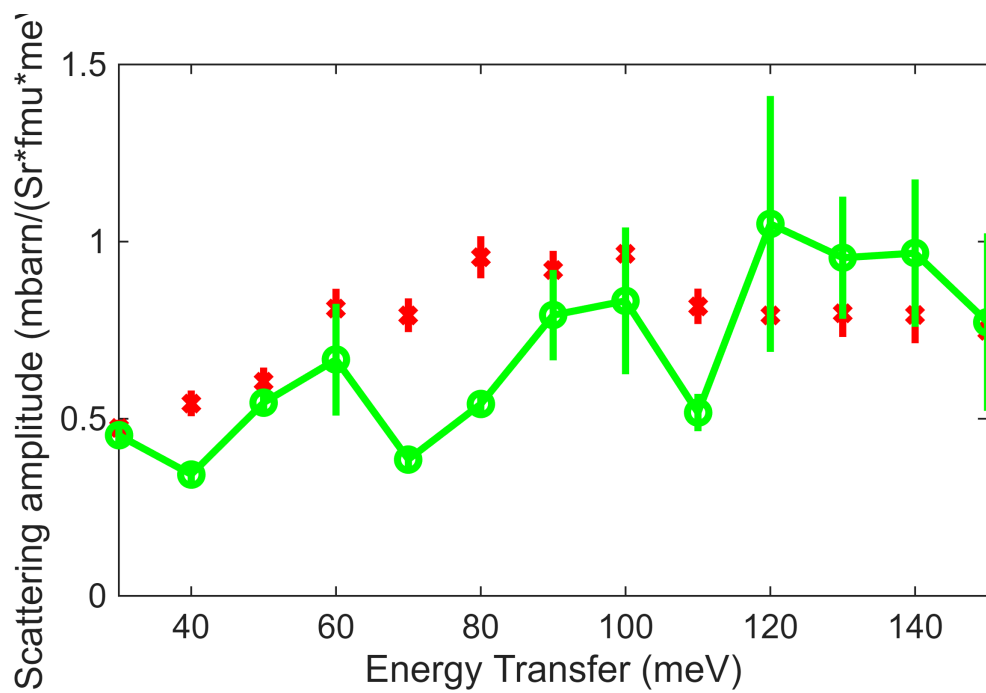
Energy (meV)	Intensity
140	-2.67
142	-2.64
144	-2.60
146	-2.79

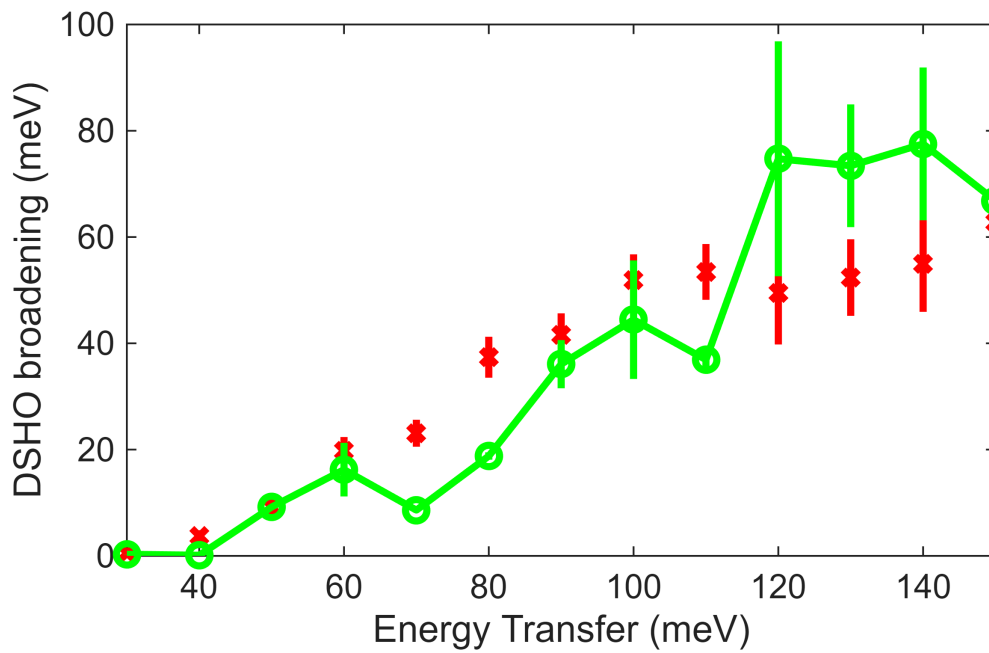
20

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 140 \leq \epsilon$$

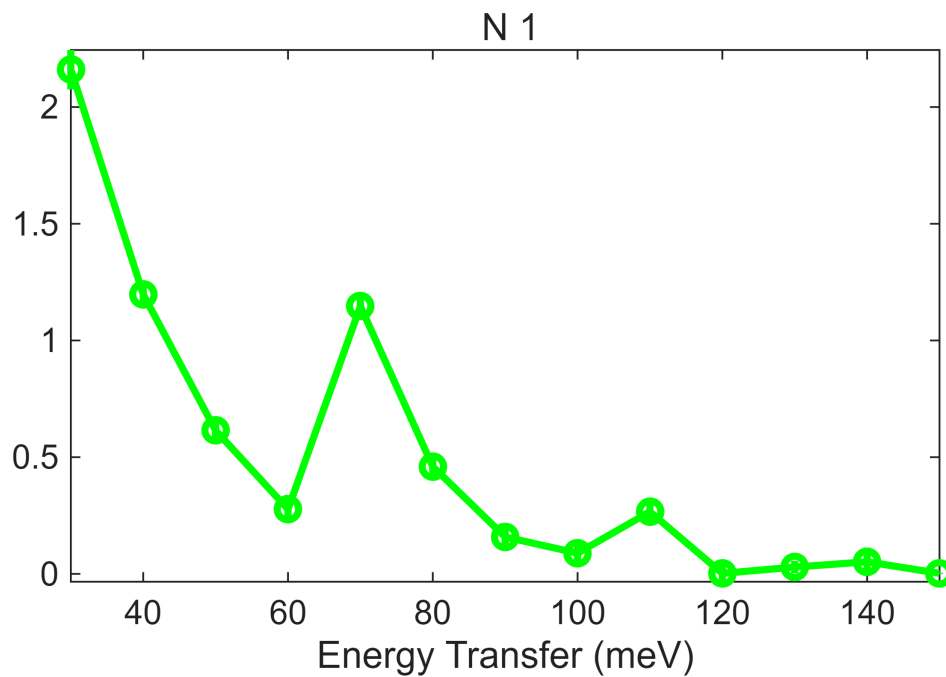
$$\zeta = -0.01:0.02:1.01 \text{ in } [2, \zeta, 0]$$

$$\xi \leq 0.1 \text{ in } [2-\xi, 0, 0], -0.1 \leq \eta \leq 0.1 \text{ in } [2, 0, \eta], 140 \leq \zeta \leq 160 \text{ in } [2, \zeta, 0]$$
[illegible]

```
plot_SJG_results(fit_res, 'g', {'S', 'J0', 'gamma'}, {[0,1.5],[20,60],[0,100]},
{Sg,J0,G});
```

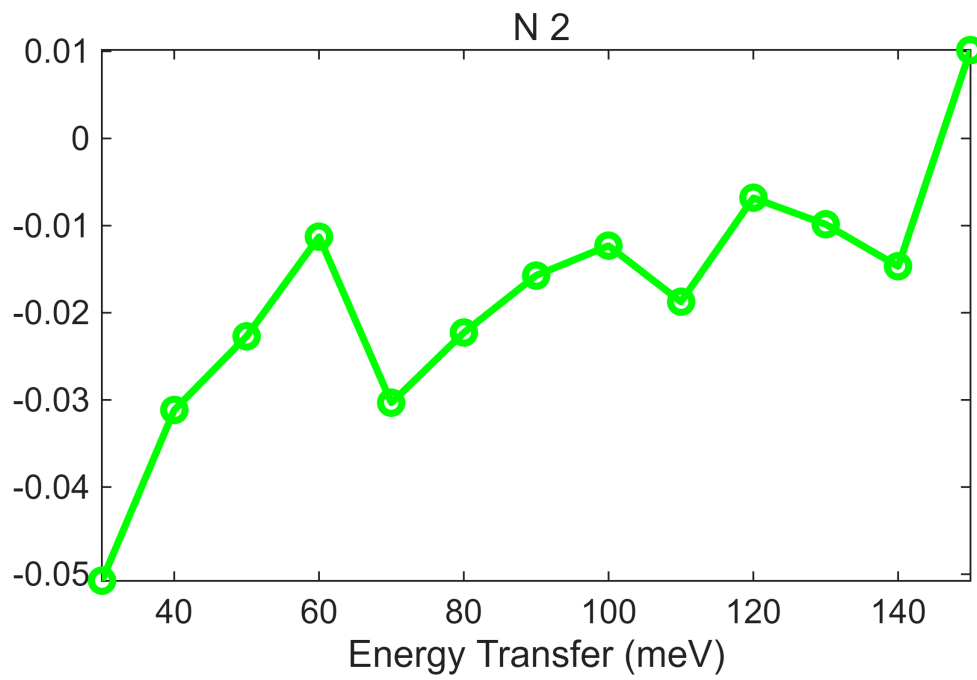




```
%bg_data = extract_bg_par(fit_res.peak_fit_par,3);
%pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_
figure;%ly -0.05 0.05
bg_data = extract_bg_par(fit_res.peak_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;%ly -0.05 0.05
```



```
pd(bg_data(2));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_f
figure;%ly -0.05 0.05
```



```

fit_res.Speak_loc_sl = Sg;
fit_res.J0peak_loc_sl = J0;
fit_res.gamma_peak_loc_sl = G;

% fit_res.Speak = Sg;
% fit_res.J0peak = J0;
% fit_res.gamma_peak = G;

fit_res.peak_fit_LoSlope_par = fit_res.peak_fit_par;
fit_resEi200 = fit_res;
save(fullfile(fit_data_path,fit_data_file),"fit_resEi200");

```